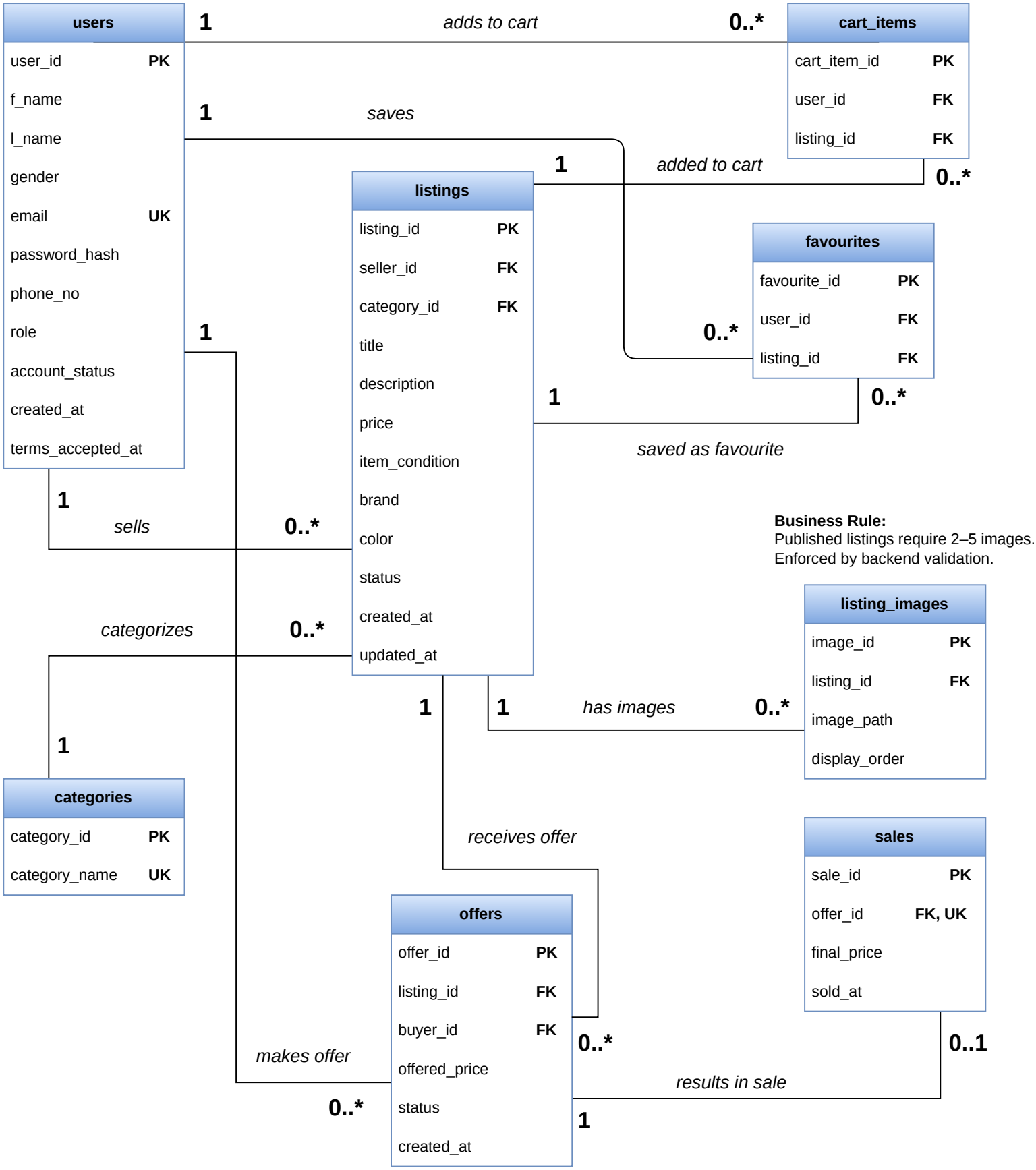
PeerMart – Entity Relationship (ER) Diagram



Business Rule:
Published listings require 2–5 images.
Enforced by backend validation.

Legend

PK - Primary Key

FK - Foreign Key

UK - Unique Key

Composite UNIQUE Key Constraints

listing_images: {listing_id, display_order}

cart_items: {user_id, listing_id}

favourites: {user_id, listing_id}

Expected Database Indexes

Index	Why is this needed?
users(email) UNIQUE	Prevents duplicate accounts with same email.
categories(category_name) UNIQUE	Prevents duplicate category names.
listing_images(listing_id, display_order) UNIQUE	Prevents duplicate image positions.
cart_items(user_id, listing_id) UNIQUE	Prevents duplicate cart entries.
favourites(user_id, listing_id) UNIQUE	Prevents duplicate favourites.
sales(offer_id) UNIQUE	Makes sure each offer has only one sale record, avoiding duplicates. Connects sales to offers, so buyer and product details can be found in users and listings.
listings(category_id, price)	Speeds up category and price filtering.
listings(created_at)	Supports newest-first sorting.
offers(buyer_id, status)	Supports buyer negotiation filtering.
offers(listing_id, status)	Supports seller negotiation filtering.